

Khush Jain

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EDUCATION

University of Wisconsin–Madison

B.S. in Computer Science & Data Science

Minor in Geospatial Data Science

Madison, WI

Sep. 2023 – May 2027

GPA: 3.6 (*Dean's List*)

Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence, Databases, Big Data Systems, Systems Design, GIS

EXPERIENCE

Undergraduate Researcher – Multimodal Foundation Models

Sep. 2026 – Present

GeoDS Lab, University of Wisconsin–Madison; Advisor: Prof. Song Gao

Madison, WI

- Developing PyTorch pipelines for multimodal data preprocessing, representation learning, and evaluation, benchmarking cross-modal alignment and geographic transferability across downstream GeoAI tasks.

Software Engineering & Geospatial Computing Intern

Aug. 2026 – Present

MultiSet AI

Madison, WI

- Implemented OGC GeoPose for 6-DoF pose data, cutting cross-platform pose-conversion latency by **35%** across cloud/WebXR/AR; built Python REST APIs for metadata validation, coordinate transforms, and standards-compliant exchange.
- Built VPS localization pipelines fusing point clouds and computer vision to detect low-feature-density regions and predict localization drift, improving 3D localization accuracy by **25%** on indoor maps.

Software Engineering & DevOps Intern

Jun. 2026 – Aug. 2026

Mass General Brigham Digital Innovation

Boston, MA

- Built and deployed a full-stack GitHub Copilot analytics platform for a **561-seat enterprise** environment using TypeScript, Vue/Nuxt, Node.js/Nitro, and GitHub REST APIs, integrating adoption, seat, billing, AI credit, model, editor, language, chat, agent, pull-request, team, and per-user telemetry.
- Built **10+ interactive analytics modules** with executive KPIs, trend visualizations, user drilldowns, cost-center attribution, license-efficiency analysis, billing-export analytics, and PostgreSQL historical reporting across **244 active developers**, **144K+ chat interactions**, **142K+ completions**, and **771K+ AI-assisted lines of code**.
- Engineered GitHub OAuth authentication, session-aware server-side RBAC, and personalized “My Usage” experiences; implemented backend data filtering, API error handling, automated tests, Docker and GitHub Actions workflows, and OpenShift deployment for the platform, which flagged **267 never-used seats** and up to **\$8K in potential monthly license waste**.

Cloud & Artificial Intelligence Intern

May 2025 – Aug. 2025

Mass General Brigham Digital Innovation

Boston, MA

- Engineered an ML-powered conversational assistant using NLP and semantic search across **2,000+ clinical workflow and policy documents**, reducing repetitive support-ticket volume by **28%**.
- Integrated intent classification and named-entity recognition, achieving **91% routing accuracy** across **15 inquiry categories** with escalation for low-confidence queries.
- Built a React UI and Python/FastAPI backend serving cited LLM responses in **under 2.5s** to **1,500+ internal pilot users**.

PROJECTS

Distributed & Cloud Data Systems | Python, Docker, gRPC, Kafka, Spark, Cassandra, HDFS, GCP

Nov 2025

- Independently built **8 end-to-end data systems** spanning containerized gRPC services, thread-safe Parquet processing, SQL-to-HDFS ETL, Spark/Hive analytics, Cassandra storage, Kafka streaming, and GCP workflows.
- Engineered fault tolerance through replication, retries, LRU caching, consistency-level tuning, checkpoint recovery, and atomic HDFS writes, recovering from service, DataNode, and consumer failures.
- Built an **8-threaded gRPC server** under a **512 MB memory limit**, processed **447K+ loan records**, and implemented a **4-partition Kafka pipeline** with crash-safe Parquet persistence.

CodeAtlas | Python, FastAPI, FAISS, MCP, RAG, Vector Embeddings

Mar 2025

- Built an end-to-end RAG assistant with document ingestion, semantic chunking, vector embedding, and FAISS indexing across **1,000+ technical documents**, serving source-grounded responses through FastAPI.
- Developed an MCP server exposing schema-validated tools and document resources for semantic search, top-*k* retrieval, and citation lookup, enabling external AI clients to invoke the retrieval pipeline programmatically.

TerraMind | Python, React, GeoPandas, PostgreSQL/PostGIS, Leaflet

May 2026

- Built a full-stack geospatial analytics application with Python REST APIs and an interactive React/Leaflet map dashboard.
- Normalized multi-source data into GeoJSON, built PostGIS spatial search/filtering, and optimized large-dataset rendering.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, JavaScript, TypeScript, C#, C, SQL, Node.js, Express.js, React, Vue/Nuxt

ML/AI: PyTorch, TensorFlow, Transformers, Fine-Tuning, LoRA, MCP, Quantization, RAG, FAISS, Prompt Engineering

Cloud & DevOps: Microsoft Azure, Docker, Linux, Bash, Git, CI/CD, Kubernetes, Terraform, Bicep

Data & Tools: REST APIs, PostgreSQL, MySQL, Pandas, NumPy, GeoPandas, Spark, Kafka, Cassandra, ArcGIS Pro